

Problem 2:

Let G be a group and let a be an element of G . Show that $g \mapsto aga^{-1}$ is a group homomorphism $G \rightarrow G$.

Proof.

Let $\varphi : G \rightarrow G$ s.t. $\varphi(g) = aga^{-1}$

Let $g, h \in G$

$$\varphi(gh) = agha^{-1} = aga^{-1}aha^{-1} = \varphi(g)\varphi(h)$$

Problem 3:

Let G and H be two groups and let $\alpha : G \rightarrow H$ and $\beta : G \rightarrow H$ be group homomorphisms. For $g \in G$, define $(\alpha\beta)(g) = \alpha(g) * \beta(g)$.

(1)

If H is commutative, show that $\alpha\beta$ is a group homomorphism.

Proof.

Assume H is commutative.

Let $g, h \in G$.

$$\begin{aligned}(\alpha\beta)(gh) &= \alpha(gh) * \beta(gh) \\&= \alpha(g) * \alpha(h) * \beta(g) * \beta(h) \\&= \alpha(g) * \beta(g) * \alpha(h) * \beta(h) = (\alpha\beta)(g) * (\alpha\beta)(h)\end{aligned}$$

(2)

Give an example of two groups and two homomorphisms such that $\alpha\beta$ is not a homomorphism.

Proof.

Let $G, H = S_3$ and $\alpha, \beta = \text{id}$.

Let $g = (12)$ and $h = (23)$.

$gh = (123)$

$$(\alpha\beta)(gh) = (123)(123) = (132) \neq e = (12)(12)(23)(23) = (\alpha\beta)(g)(\alpha\beta)(h)$$

Problem 4:

Let G and H be two groups. We define the **product group** $G \times H$ to have, as underlying set, the Cartesian product $G \times H$, with operation

$$(g_1, h_1)(g_2, h_2) = (g_1g_2, h_1h_2).$$

Remark.

Define $\Delta : G \rightarrow G \times G$ by $\Delta(g) = (g, g)$, and define $\nabla : G \rightarrow G \times G$ by $\nabla(g, h) = gh$.

(2)

Is Δ always a group homomorphism?

Proof.

Yes, let $g, h \in G$.

$$\begin{aligned}\Delta(gh) &= (gh, gh) \\ &= (g, g)(h, h) = \Delta(g)\Delta(h)\end{aligned}$$

(3)

Is ∇ always a group homomorphism?

Proof.

No, let $G = S_3$ and $g = (12), h = (23)$.

$$\nabla((g, g)(h, h)) = \nabla((gh, gh)) = ghgh = (132) \neq e = gghh = \nabla(g, g)\nabla(h, h)$$

Problem 5:

Let S_n act on $\mathbb{Q}[x_1, x_2, \dots, x_n]$ in the obvious way. Let $\Delta = \prod_{1 \leq i < j \leq n} (x_i - x_j) \in \mathbb{Q}[x_1, x_2, \dots, x_n]$. For $\sigma \in S_n$, let $M(\sigma)$ be the corresponding permutation matrix in $\text{GL}_n(\mathbb{Q})$.

(1)

Show that

$$\frac{\sigma(\Delta)}{\Delta} = (-1)^{|\{(i,j) | 1 \leq i < j \leq n, \sigma(i) > \sigma(j)\}|} = \det M(\sigma)$$

Proof.

$$\begin{aligned} \sigma(\Delta) &= \prod_{1 \leq i < j \leq n} (x_{\sigma(i)} - x_{\sigma(j)}) \\ x_{\sigma(i)} - x_{\sigma(j)} &= \begin{cases} x_i - x_j, & \sigma(i) < \sigma(j), \\ -(x_i - x_j), & \sigma(i) > \sigma(j). \end{cases} \end{aligned}$$

Therefore,

$$\sigma(\Delta) = (-1)^{|\{(i,j) | 1 \leq i < j \leq n, \sigma(i) > \sigma(j)\}|} \Delta \implies \frac{\sigma(\Delta)}{\Delta} = (-1)^{|\{(i,j) | 1 \leq i < j \leq n, \sigma(i) > \sigma(j)\}|}$$

A permutation matrix is made by swapping rows. The determinant is 1 if there is an even number of swaps and -1 if there is a odd number.

Thus,

$$\det(M(\sigma)) = (-1)^{|\{(i,j) | 1 \leq i < j \leq n, \sigma(i) > \sigma(j)\}|}$$

(2)

We defined $\epsilon(\sigma)$ to be given by any of the formulas. Show that ϵ is a group homomorphism $S_n \rightarrow \{\pm 1\}$.

Proof.

Let $\sigma, \tau \in S_n$.

$$\epsilon(\sigma\tau) = \det(M(\sigma\tau)) = \det(M(\sigma)M(\tau)) = \det(M(\sigma))\det(M(\tau)) = \epsilon(\sigma)\epsilon(\tau)$$

Remark.

- The **alternating group** A_n is defined as the kernel of ϵ .

(3)

Show that S_n is generated by the elements $(1j)$ for $2 \leq j \leq n$.

Proof.

Consider k -cycle $\sigma = (a_1 a_2 \dots a_k)$.

Consider $(1a_1)(1a_k)(1a_{k-1}) \dots (1a_2)(1a_1)$

Let $i \in \{1, \dots, n\}$.

If $i = a_j$ for $j \in \{1, \dots, k\}$ then

$$(1a_1)(1a_k)(1a_{k-1}) \dots (1a_2)(1a_1)(a_j) = a_{j+1 \bmod k} = \sigma(a_j)$$

If $i \neq a_j$ for $j \in \{1, \dots, k\}$ then

$$(1a_1)(1a_k)(1a_{k-1}) \dots (1a_2)(1a_1)(i) = i = \sigma(i)$$

Thus, we can make any k cycle out of $(1j)$ for $2 \leq j \leq n$. Thus, we can make any element in S_n . ■

(4)

Show that A_n is generated by the elements $(12k)$ for $3 \leq k \leq n$.

Proof.

Let $\sigma \in A_n$. From (3) we can write $\sigma = (1a_1)(1a_2) \dots (1a_k)$. Since we define $A_n = \ker(\epsilon)$.

$$\epsilon(\sigma) = \epsilon(1a_1)\epsilon(1a_2) \dots \epsilon(1a_k) = 0$$

Since $\epsilon(1a_i) = -1$ this implies k is even.

This implies that A_n when decomposed has an even amount of transpositions. Thus, we only need to look at $(1a)(1b)$ and generate it with $(12k)$.

If $a = b$: $(1a)(1b) = e$

If $a \neq b$:

If $a, b \geq 2$: $(1a)(1b) = (12a)(12b)(12b)$

If $a = 2$: $(12)(1b) = (12b)(12b)$

If $b = 2$: $(1a)(12) = (12a)$

In every case, $(1a)(1b)$ is a product of $(12k)$ with $k \geq 3$, and since every $\sigma \in A_n$ can be written with an even amount of transpositions of the form $(1a)$. This implies that A_n is generated by the elements $(12k)$ for $k \geq 3$. ■

Problem 6:**(1)**

Let n be a positive integer and a another integer. Show that there exists x s.t. $ax \equiv 1 \pmod n$ if and only if $\gcd(a, n) = 1$.

Proof. $(\implies)$:

Assume $\exists x$ s.t. $ax \equiv 1 \pmod n$.

This means there exists a $y \in \mathbb{Z}$ s.t.

$$ax - 1 = ny \implies ax - ny = 1$$

Let $d = \gcd(a, n)$. Since $d \mid a$ and $d \mid n$ this implies $d \mid 1$. Thus, $\gcd(a, n) = d = 1$.

 $(\impliedby)$:

Assume $\gcd(a, n) = 1$.

By Bezout's Lemma there exists $x, y \in \mathbb{Z}$ s.t. $ax + ny = 1$. Thus $ax - 1 = -ny \implies ax \equiv 1 \pmod n$. Thus, there exists an x . ■

(2)

Let $(\mathbb{Z}/n\mathbb{Z})^\times$ be the set of residue classes $\bar{a}$ modulo n for which $\gcd(a, n) = 1$. Show that $(\mathbb{Z}/n\mathbb{Z})^\times$ is a group.

Proof.

Associativity:

It is by definition associative.

Identity element:

$1 \in (\mathbb{Z}/n\mathbb{Z})^\times$ and for all $g \in (\mathbb{Z}/n\mathbb{Z})^\times$ $1g = g = g1$.

Inverse element:

Let $g \in (\mathbb{Z}/n\mathbb{Z})^\times$. This means that $\gcd(g, n) = 1$. Thus, by (1) there exists an $x \in \mathbb{Z}$ s.t. $gx \equiv 1 \pmod n$. Therefore, $g\bar{x} = 1 = \bar{x}g$.

Thus, $(\mathbb{Z}/n\mathbb{Z})^\times$ is a group. ■

(3)

Let n have prime factorization $n = p_1^{a_1} p_2^{a_2} \dots p_r^{a_r}$. Define a map $(\mathbb{Z}/n\mathbb{Z})^\times \rightarrow \prod (\mathbb{Z}/p_i^{a_i}\mathbb{Z})^\times$ by $(m \bmod p_1^{a_1}, m \bmod p_2^{a_2}, \dots, m \bmod p_r^{a_r})$, where the product of groups is defined in. Show that this map is an isomorphism of groups.

Proof.

Let φ be that map.

Well-Defined:

Let $m \equiv m' \pmod n$, then since $p_i^{a_i} \mid n$, we have

$$m \equiv m' \pmod{p_i^{a_i}}$$

for every i . Hence, each coordinate is well-defined.

Homomorphism:

Let $g, h \in (\mathbb{Z}/n\mathbb{Z})^\times$.

$$\begin{aligned} \varphi(gh) &= (gh \bmod p_1^{a_1}, gh \bmod p_2^{a_2}, \dots, gh \bmod p_r^{a_r}) \\ &= (g \bmod p_1^{a_1}, g \bmod p_2^{a_2}, \dots, g \bmod p_r^{a_r}) \\ &\quad \cdot (h \bmod p_1^{a_1}, h \bmod p_2^{a_2}, \dots, h \bmod p_r^{a_r}) \\ &= \varphi(g)\varphi(h) \end{aligned}$$

Injective:

Let $g \in \ker(\varphi)$. Then

$$g \equiv 1 \pmod{p_i^{a_i}}$$

Since $p_i^{a_i}$ are pairwise coprime, the Chinese remainder theorem gives $g \equiv 1 \pmod n$.

Thus, $g = 1$.

Surjective:

Let $(g_1, g_2, \dots, g_r) \in \prod (\mathbb{Z}/p_i^{a_i}\mathbb{Z})^\times$

By the Chinese remainder theorem, there exists an integer m s.t.

$$m \equiv g_i \pmod{p_i^{a_i}}$$

Thus, $\varphi(m) = (g_1, g_2, \dots, g_r)$. Therefore, φ is an isomorphism, and

$$(\mathbb{Z}/n\mathbb{Z})^\times \cong \prod (\mathbb{Z}/p_i^{a_i}\mathbb{Z})^\times$$